

ERT Field Notes — Whiteboard

1. Why do we measure voltage multiple times?

The instrument injects current and measures the resulting voltage — but the reading is never perfectly stable. Noise from the ground, cables, and electronics causes small fluctuations. To deal with this, we repeat the measurement 5 times at each point.

Example: 5 voltage readings at one point

Measurement	V (mV)
1	112
2	108
3	115
4	108
5	111

2. Best estimate → Mean

We take the average as our best estimate of the true voltage:

$$\bar{V} = (112 + 108 + 115 + 108 + 111) / 5 = 554 / 5 = 110.8 \text{ mV}$$

3. Spread of the data → Standard Deviation (σ)

The standard deviation tells us how far each measurement deviates from the mean:

$$\sigma = \sqrt{[\Sigma (V_i - \bar{V})^2 / n]} \approx 2.5 \text{ mV}$$

4. Final reported value

$$V = 110.8 \pm 2.5 \text{ mV}$$

- 110.8 mV → our best estimate (mean)
- ± 2.5 mV → uncertainty (standard deviation)
- Smaller σ = more reliable measurement

Key idea: The instrument never gives the same value twice. The mean is our best estimate; σ tells us how much we can trust it.

Lateral vs. Vertical Resolution

Resolution is the ability to distinguish two separate features. In ERT there are two directions to consider:

Lateral Resolution	Vertical Resolution
• Ability to distinguish two objects side by side • Controlled by electrode spacing (a) • Smaller spacing $\rightarrow$ better lateral resolution • Dipole-Dipole: best lateral resolution	• Ability to distinguish layers at different depths • Controlled by how deep current penetrates • Larger spacing (a) $\rightarrow$ greater depth • Wenner & Schlumberger: best vertical resolution

Simple analogy

Think of electrode spacing (a) as the 'zoom level' of your survey:

- Small $a \rightarrow$ zoomed in, see fine detail sideways (lateral) but only shallow
- Large $a \rightarrow$ zoomed out, reach greater depth (vertical) but lose fine lateral detail

Which array wins?

Array	Lateral resolution	Vertical resolution
Wenner	Moderate	Good
Schlumberger	Lower	Very good
Dipole-Dipole	Best	Weaker signal

There is always a trade-off: higher lateral resolution comes at the cost of weaker signal and less depth penetration.